

A Measurement Model of Cognitive Flexibility

Dave Braun

Background

Flexibility is a hallmark of cognition and is typically measured by examining features of task data. For example, in task switching, lower switch costs between tasks are generally taken to reflect higher levels of cognitive flexibility. In voluntary task switching, higher voluntary switch rates are also taken to reflect higher cognitive flexibility. However, summarizing these task metrics with traditional techniques (ie, t tests, ANOVAs) forces researchers to focus on either switch costs *or* switch rates (while often ignoring error rates and individual differences entirely).

An ideal measure of cognitive flexibility would take all observed data into account simultaneously while characterizing how flexibility differs between individuals. Such integration across observed data to estimate latent variables of interest is one of the many advantages of computationally modeling cognitive processes. Here, I propose a model of behavior in a hybrid cued/voluntary task switching environment, explained in terms of parameters related to cognitive flexibility.

Task and effects

The hybrid task-switching paradigm (Frober, Jurczyk, & Dreisbach, 2022; *JEP:LMC*) interleaves cued and voluntary task switching trials within a block, where the number of stimuli presented on the screen signals whether it is a forced (single stimulus) or free (two stimuli) trials. Experimentally manipulated variables include the ratio of free to forced trials (forced switch rate), whether a trial is cued or voluntary (trial type), and whether the task was repeated or switched across a trial (transition). On each trial there are three dependent variables: task choice, task accuracy, and response time (RT).

The core phenomenon of interest in this task is that switch costs decrease and voluntary switch rates increase after a forced switch trial. Frober et al. (2022) attribute this to *switching-induced flexibility*, meaning that a forced task switch induces increased cognitive flexibility that transfers to a voluntary task environment. This boost in cognitive flexibility is thought to decay as more voluntary trials are performed, as evidenced by reduced voluntary switch rate.

Model

I propose a joint model of task choice, task accuracy, and response time in the hybrid task-switching paradigm, explained in terms of various aspects of cognitive flexibility that can vary within and across people. This model integrates task choice and execution processes by structurally binding parameters across these processes. This model is focused more on measuring different components of cognitive flexibility rather than providing a mechanistically precise account of cognitive flexibility (as in eg a dynamical systems model).

Task Choice

A choice on trial i for participant j is a function of how many trials it has been since a forced switch trial (*SFS*):

$$\begin{aligned}\text{logit}(\omega_{ij}) &= \phi_j + \beta_j e^{-\lambda_j \cdot \text{SFS}_{ij}} \\ \text{Ch}_{ij} &\sim \text{Bern}(\omega_{ij})\end{aligned}$$

Where the log odds of the probability of switching tasks ($\text{logit}(\omega_{ij})$) for participant i on trial j is a decay function with: ϕ_j reflecting individual differences in baseline cognitive flexibility, β_j reflecting individual differences in the degree of boost to cognitive flexibility after a forced switch, and λ_j reflecting individual differences in how quickly that boost to flexibility decays over trials. The probability of switching tasks (ω_{ij}) parameterizes a Bernoulli distribution to map to the task transition (Ch_{ij}) on trial i for participant j . $\text{SFS}_{ij} = 0$ when trial type is a forced switch, leading the exponential term to reduce to 1, which increases the baseline cognitive flexibility by β_j . As SFS_{ij} increases, the exponential approaches zero, diminishing the impact of β_j on the log odds of switching.

Task Performance

Task accuracy and response time are modeled as a diffusion process, where the drift rate for switching vs. repeating tasks is influenced by the probability of switching tasks (ω_{ij}) on the *previous* trial.

$$\begin{aligned}\nu_{0,i,j} &= \gamma_{0,j} \cdot \omega_{i,j-1} \\ \nu_{1,i,j} &= \gamma_{1,j} \cdot \omega_{i,j-1}\end{aligned}$$

Where individual differences in repeat ($\gamma_{0,j}$) and switch ($\gamma_{1,j}$) drift rates are scaled by the probability of switching tasks on the previous trial, to obtain repeat ($\nu_{0,i,j}$) and switch ($\nu_{1,i,j}$) rates for participant i on trial j . A trial can only be a switch *or* repeat, so we use the observed task choice ($\text{Ch}_{ij} \in \{0 = \text{repeat}, 1 = \text{switch}\}$) to determine which drift gets implemented:

$$V_{ij} = \nu_{0,i,j} \cdot (1 - \text{Ch}_{ij}) + \nu_{1,i,j} \cdot \text{Ch}_{ij}$$

Finally, the task accuracy and RT on trial i for participant j are modeled with the DDM

$$\{\text{Acc}_{ij}, \text{RT}_{ij}\} \sim \text{DDM}(V_{ij}, \alpha_j, \tau_j, \sigma_j)$$

Where α_j , τ_j , and σ_j reflect individual differences in boundary separation, non-decision time, and variability in drift rate, respectively.

ω as a within-person measure of state cognitive flexibility

By binding ω_{ij} to the choice and performance processes, I intend for it to become a within-person measure of cognitive flexibility across trials. As ω_{ij} goes towards 1 (more flexible), switch rate increases and switch costs decrease. As it goes towards 0 (more stable), switch rates decrease and switch costs increase. Binding drift rates to the cognitive state on trial $j - 1$ captures the time needed to reconfigure / reconcile persisting activation of task sets.

Aspects of cognitive flexibility

This model in theory measures four aspects of cognitive flexibility: three between-person measures, and one-within person measure.

Between-person

- ϕ_j : Individual differences in baseline cognitive flexibility
- β_j : Individual differences in the boost to cognitive flexibility from a forced switch
- λ_j : Individual differences in the rate at which the boost in flexibility decays after a forced switch

Within-person

- ω_{ij} : A normalized (between 0 and 1) measure of state cognitive flexibility

This strikes me as an exciting opportunity to assess the degree to which these parameters are associated with covariates. For example, I might expect ϕ_j to correlate with differences in age, and β_j to correlate to differences in reward sensitivity. It would be extra exciting if these parameters explained more variance in these covariates than typical task metrics (eg, switch costs). There are opportunities to associate ω_{ij} with simultaneously collected data, such as neural measures or thought probes.

Fitting the data

Two of the central effects reported in Frober et al (2022) are that (1) switch costs are greater on forced rather than free choice trials, and (2) the voluntary switch rate following a forced choice trial is greater than that following a free choice trial. Simulations from the proposed model can capture both of these effects:

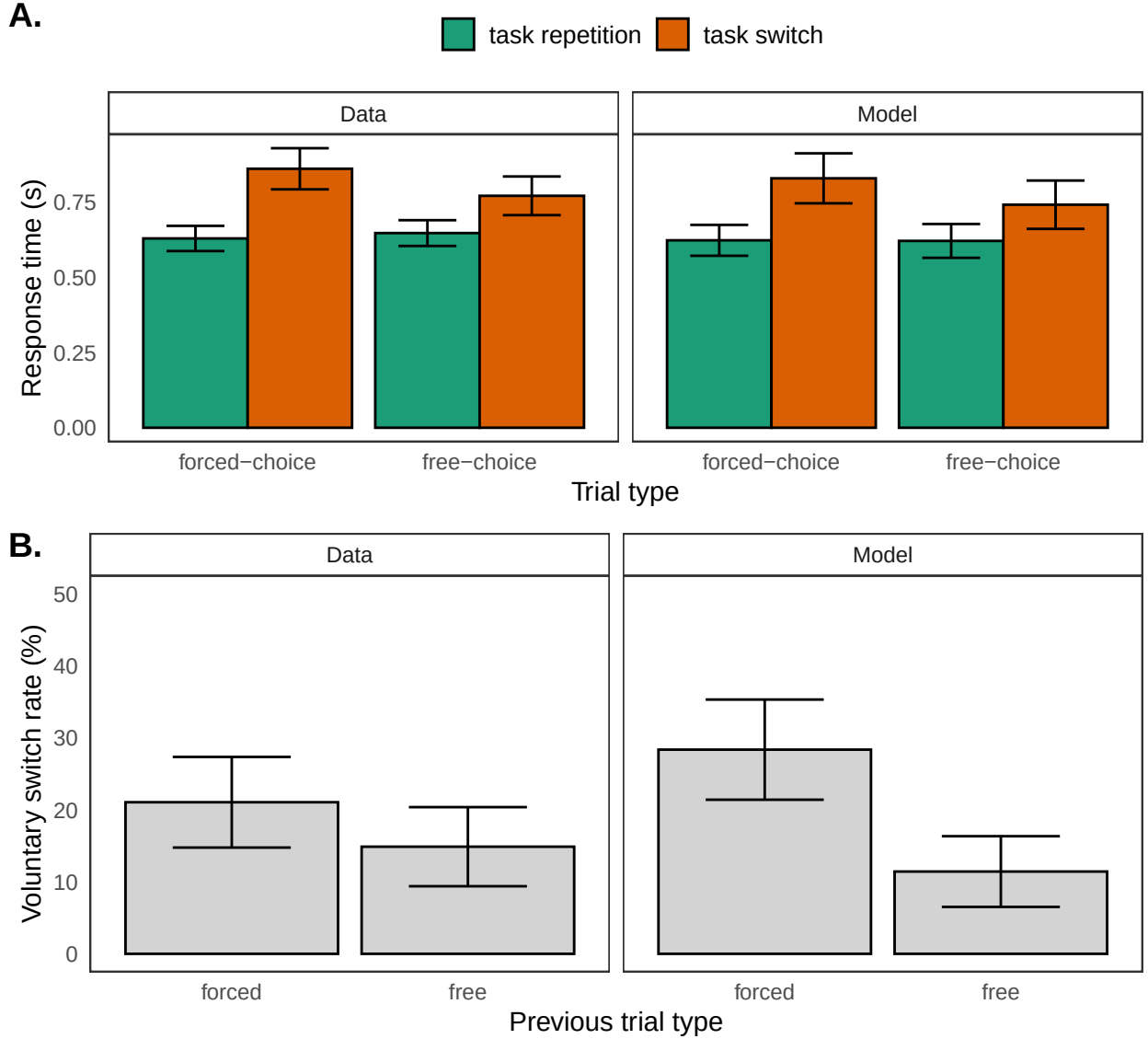


Figure 1: Fitting the model to data from Frober et al. (2022; Experiment 2). (A) Response time (s) as a function of task transition and trial type. (B) Voluntary switch rate as a function of previous trial type. Model is fit to data with hand-tuned parameters.